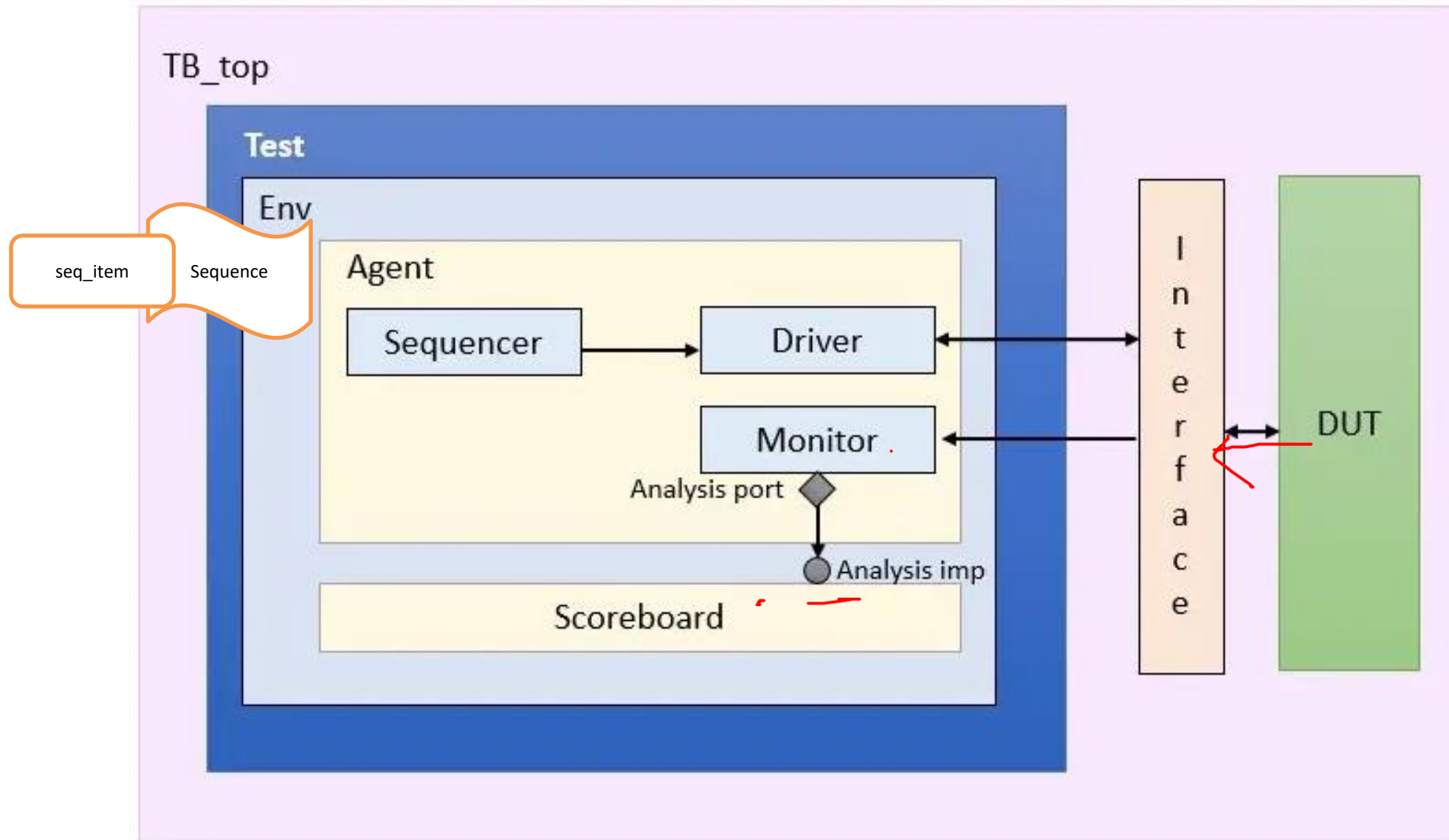


Using DPI-C

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Recall the UVM based Testbench



SystemVerilog Direct Programming Interface (DPI)

- What is DPI?
- Enables communication between SystemVerilog and C/C++
- Lightweight and efficient foreign language interface
- Use case in UVM:
 - To compare C- based Golden reference model in scoreboard
-  **Import functions:**
 - Call C functions from SystemVerilog
- **Export functions:**
 - Call SystemVerilog functions/tasks from C

Importing C Functions

- Syntax:
 - import "DPI-C" function <return_type> <function_name> (<arguments if any>)
- Call in SV just like a normal function
- <https://www.edaplayground.com/x/iHMF>

Exporting C Functions

- Syntax:
 - export "DPI-C" function <function_name>
- Useful when C code needs callbacks into the simulation
- <https://www.edaplayground.com/x/iHMF>
- <https://www.edaplayground.com/x/Uqy3>

Using C based Model in Scoreboard

- <https://www.edaplayground.com/x/X7ee>